High-Dimensional Statistics: A Non-Asymptotic Viewpoint

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Spring 2024







Chapter I
Introduction

Part 1
Linear Discriminant Analysis

Linear Discriminant Analysis (LDA)

Objective of LDA:

- To determine whether an observed vector $\mathbf{x} \in \mathbb{R}^d$ is drawn from one of two possible distributions (e.g., $\mathbb{P}_1$ or $\mathbb{P}_2$).

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- Allows trade-off between two error types—incorrectly deciding $\mathbb{P}_1$ when $\mathbb{P}_2$ is true (Type I error), and vice versa (Type II error).

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Neyman–Pearson Lemma:

- Provides an optimal family of decision rules through randomization, balancing Type I and Type II errors.

Evaluating the Quality of Decision Rule via Error Probability

Gaussian Case:

- For multivariate Gaussian distributed classes with different mean vectors μ_1, μ_2 and common covariance Σ, the ratio $\log \frac{\mathbb{P}_1[\mathbf{x}]}{\mathbb{P}_2[\mathbf{x}]}$ simplifies to:

$$\Psi(x) := (\mu_1 - \mu_2)^\top \Sigma^{-1} \left(x - \frac{\mu_1 + \mu_2}{2} \right)$$

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Error Probability:

- Assume X' and X'' are random vectors from $\mathbb{P}_1$ and $\mathbb{P}_2$ respectively. Then the Error Probability is:

$$\text{Err}(\Psi) = \frac{1}{2}\mathbb{P}_1[\Psi(X') \leq 0] + \frac{1}{2}\mathbb{P}_2[\Psi(X'') > 0]$$

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Under the Multivariate Gaussian Distribution Assumption:

$$\text{Err}(\Psi) = \underbrace{\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{-\gamma/2} e^{-t^2/2} dt}_{\Phi(-\gamma/2)}, \text{ where } \gamma = \sqrt{(\mu_1 - \mu_2)^\top \Sigma^{-1} (\mu_1 - \mu_2)}$$

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Assumption: Sample covariance matrix is invertible ($n_i > d$).

- The discriminant function is based on sample estimates.

$$\widehat{\Psi}(x) = \left\langle \hat{\mu}_1 - \hat{\mu}_2, \widehat{\mathbf{\Sigma}}^{-1} \left(x - \frac{\hat{\mu}_1 + \hat{\mu}_2}{2} \right) \right\rangle .$$

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$$\hat{\Sigma} := \frac{1}{n_1-1}\sum_{i=1}^{n_1}(x_i-\hat{\mu}_1)(x_i-\hat{\mu}_1)^\top + \frac{1}{n_2-1}\sum_{i=n_1+1}^{n_1+n_2}(x_i-\hat{\mu}_2)(x_i-\hat{\mu}_2)^\top$$

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- Assume classes are equally likely, and the error probability with zero threshold:

$$\mathrm{Err}(\widehat{\Psi}) := \frac{1}{2}\mathbb{P}_1\left(\widehat{\Psi}\left(X'\right) \le 0\right) + \frac{1}{2}\mathbb{P}_2\left(\widehat{\Psi}\left(X''\right) > 0\right)$$

Practical Linear Discriminant Analysis (Cont.)

Kolmogorov's Simplification (1960):

- Assume $\boldsymbol{\Sigma}$ is the identity $\quad \Rightarrow \quad \widehat{\Psi}_{\mathrm{id}}(x) = \left\langle \hat{\mu}_1 - \hat{\mu}_2, x - \frac{\hat{\mu}_1 + \hat{\mu}_2}{2} \right\rangle$

High-Dimensional Asymptotics:

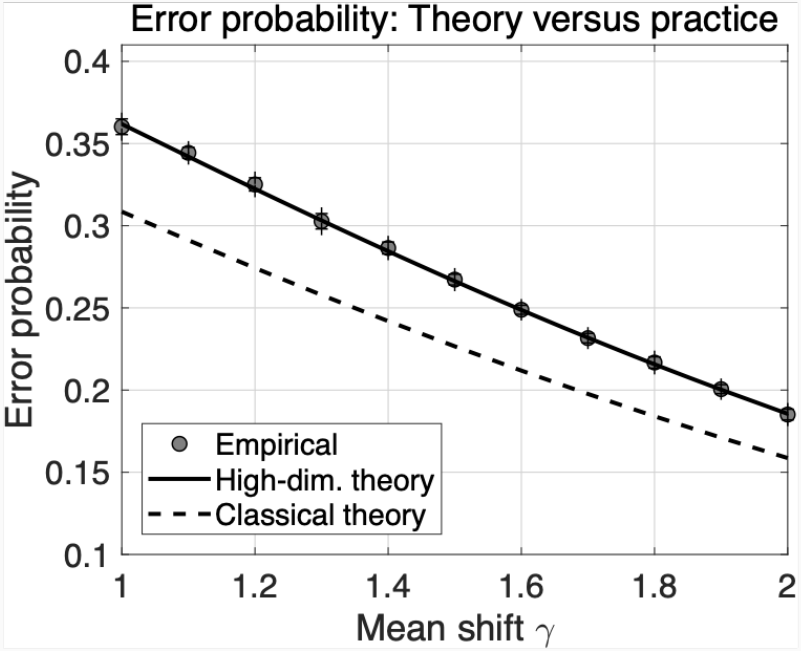
- Consider the behavior under Gaussian assumption with high-dimensional data $(n_1, n_2, d \to \infty)$.
- Ratios d/n_i converge to a non-negative fraction $\alpha > 0$.
- The Euclidean distance $||\mu_1 - \mu_2||$ converges to a constant $\gamma > 0$.

Asymptotic Error Probability:

- $\mathrm{Err}(\Psi_{\mathsf{id}})$ converges in probability to a fixed number.
- Classical scaling prediction is modified in high dimensions.

$$\mathrm{Err}(\Psi_{\mathsf{id}}) \xrightarrow{\text{prob.}} \Phi\left(-\frac{\gamma^2}{2\sqrt{\gamma^2 + 2\alpha}}\right)$$

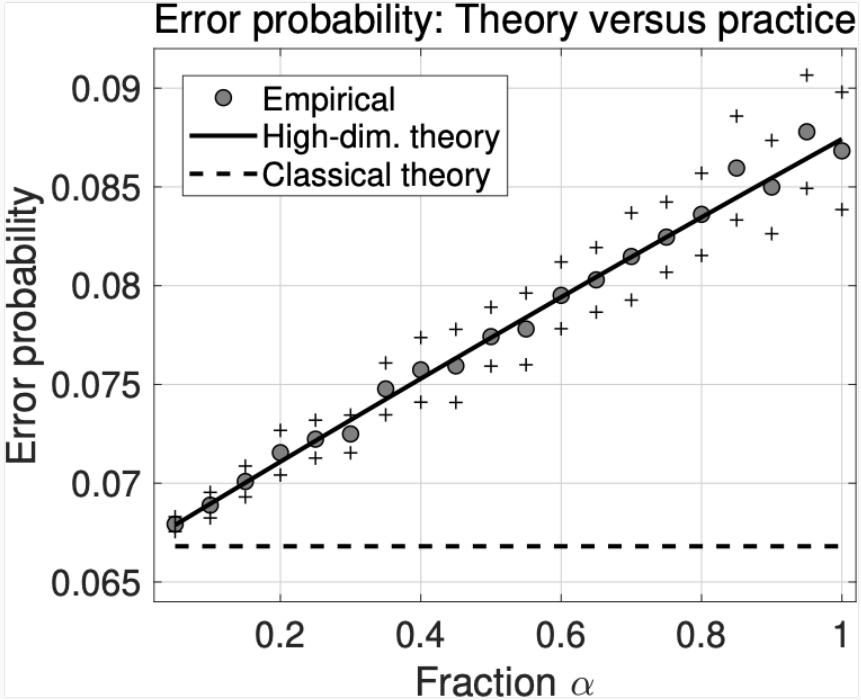
Practical Linear Discriminant Analysis (Cont.)



(a)

- Error probability $Err(\hat{\Psi}_{id})$ vs. mean shift parameter $\gamma \in [1, 2]$
  - Parameters: $d = 400$, $\alpha = 0.5$, $n_1 = n_2 = 800$
  - Empirical error probabilities (gray circles) averaged over 50 trials
  - High-dimensional prediction (solid curve)
  - Classical prediction (dashed curve)
  - The Empirical and Classical theory do not match

Practical Linear Discriminant Analysis (Cont.)

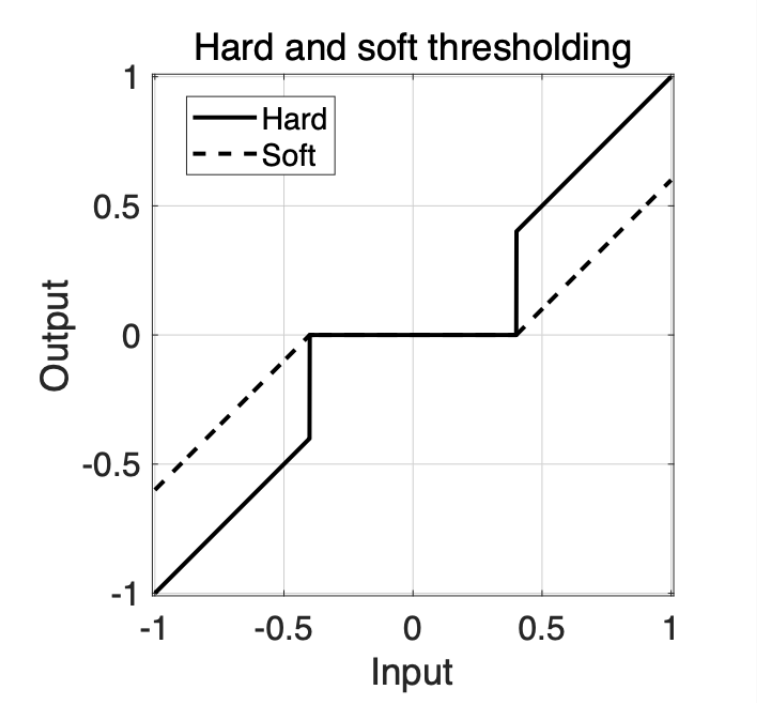


(b)

- Error probability $Err(\hat{\Psi}_{id})$ vs. fraction $\alpha \in [0, 1]$
  - Parameters: $d = 400$, $\gamma = 2$
  - Classical prediction (dashed line) remains flat, independent of α

Solution to match the Empirical and Classical theory

Solution: Thresholding the sample means using *e.g.* soft/hard thresholding:

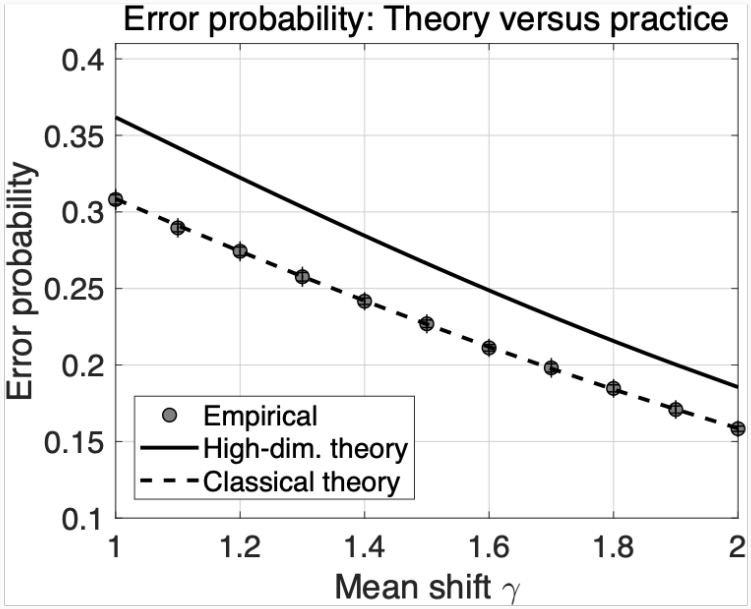


$$H_\lambda(x) = x1[|x| > \lambda] = \begin{cases} x & \text{if } |x| > \lambda, \\ 0 & \text{otherwise} \end{cases}$$

$$T_\lambda(x) = 1[|x| > \lambda](x - \lambda\,\mathrm{sign}(x)) = \begin{cases} x - \lambda\,\mathrm{sign}(x) & \text{if } |x| > \lambda, \\ 0 & \text{otherwise .} \end{cases}$$

Solution to match the Empirical and Classical theory (Cont.)

- **Threshold**: $\lambda_n := \sqrt{\frac{2 \log d}{n}}$
- **Thresholded Means**: $\widetilde{\mu}_j = H_{\lambda_n}(\hat{\mu}_j)$ for $j = 1, 2$
- **Standard Tail Bounds**: Justification for λ_n choice is provided in Chapter 2 (see Exercise 2.12).
- **Linear Discriminant**: $\widetilde{\Psi}(x) := \left\langle \widetilde{\mu}_1 - \widetilde{\mu}_2, x - \frac{\widetilde{\mu}_1 + \widetilde{\mu}_2}{2} \right\rangle$
- **Results**: Error rate $\mathrm{Err}(\widetilde{\Psi})$ depicted below:



(b)

- The Empirical and Classical theory are matched now.

Summary

- **Linear Discriminant Analysis (LDA)**
  - *Objective*: Classify vectors from two distributions using log-likelihood ratio.
  - *Decision Rule*: Thresholding to balance Type I and II errors.
  - *Neyman-Pearson Lemma*: Optimal decision rules through randomization.
- **Evaluating Decision Rules**
  - *Gaussian Case*: Log-likelihood ratio simplifies with Gaussian distributions.
  - *Error Probability*: Calculated under multivariate Gaussian assumption.
- **Practical LDA**
  - *Estimation*: Calculate sample means and covariance matrices.
  - *Discriminant Function*: Based on sample estimates with assumptions on sample covariance invertibility.
- **High-Dimensional Asymptotics**
  - *Asymptotic Error Probability*: Modified classical scaling predictions in high dimensions.

Part 2
Covariance estimation

Covariance estimation

Objective of covariance estimation:

- Let $x_1, \ldots, x_n$, where x_i is i.i.d from a zero-mean distribution in $\mathbb{R}^d$. We want to estimate $\mathbf{\Sigma} = \mathrm{cov}(X)$

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- One useful matrix norm is the ℓ_2-operator norm, given by

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- Then with fixed d, $\|\widehat{\mathbf{\Sigma}} - \mathbf{\Sigma}\|_2 \underset{n \to +\infty}{\longrightarrow} 0$ (a.s). What about when the ratio $d/n = \alpha \in (0, 1)$ stays fixed?

Covariance estimation

Simulation:

- Let $\boldsymbol{\Sigma} = \boldsymbol{I}_d$ with each $X_i \sim \mathcal{N}(0, \boldsymbol{I}_d)$ for each $i = 1, \ldots, n$.

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- Using these n samples, we generated $\widehat{\boldsymbol{\Sigma}}$, and then computed its vector of eigenvalues $\gamma(\widehat{\Sigma}) \in \mathbb{R}^d$ as

$$\gamma_{\max}(\widehat{\boldsymbol{\Sigma}}) = \gamma_1(\widehat{\boldsymbol{\Sigma}}) \geq \gamma_2(\widehat{\boldsymbol{\Sigma}}) \geq \cdots \geq \gamma_d(\widehat{\boldsymbol{\Sigma}}) = \gamma_{\min}(\widehat{\boldsymbol{\Sigma}}) \geq 0$$

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Theory:

- The eigenvalues distribution follows the Marčenko-Pastur law $f_{\mathrm{MP}}(\gamma) \propto \sqrt{\frac{(t_{\max}(\alpha)-\gamma)(\gamma-t_{\min}(\alpha))}{\gamma}}$, supported on the interval $[t_{\min}(\alpha), t_{\max}(\alpha)] = \left[(1-\sqrt{\alpha})^2, (1+\sqrt{\alpha})^2\right]$. α is called the aspect ratio and is $\alpha = d/n$.

Covariance estimation - simulation

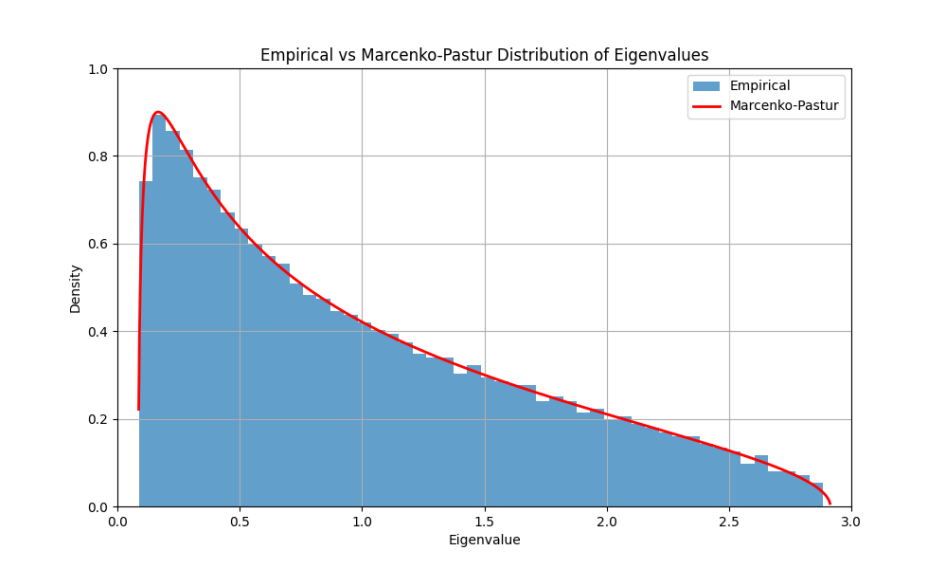


Figure 1: Empirical distribution of the eigenvalues of a sample covariance matrix $\widehat{\mathbf{\Sigma}}$ versus the asymptotic prediction of the Marčenko-Pastur law. Aspect ratio $\alpha = 0.5$

Covariance estimation - simulation

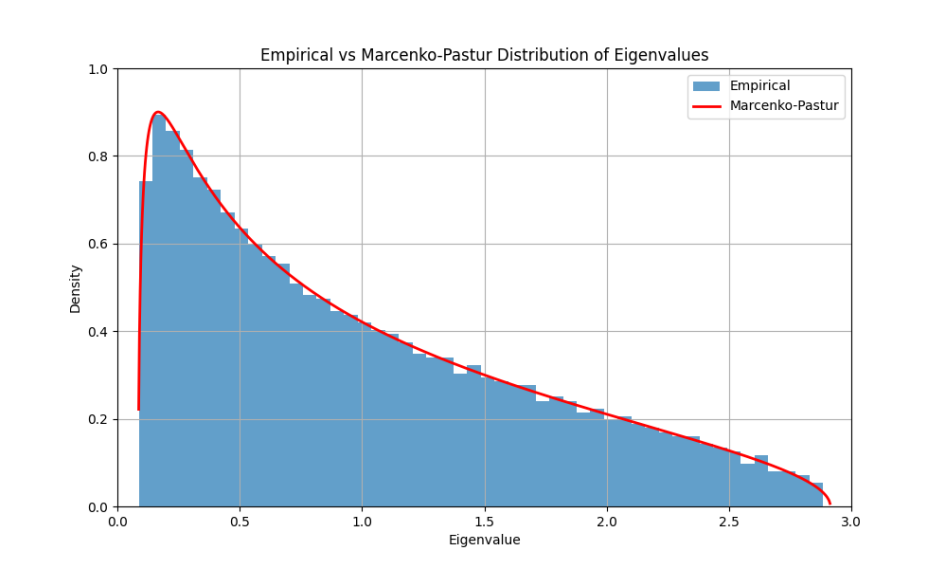


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Problem: Even with large n, so long as α is fixed, the eigenvalues do not converge to 1!

Covariance estimation - simulation

Problem: Yet

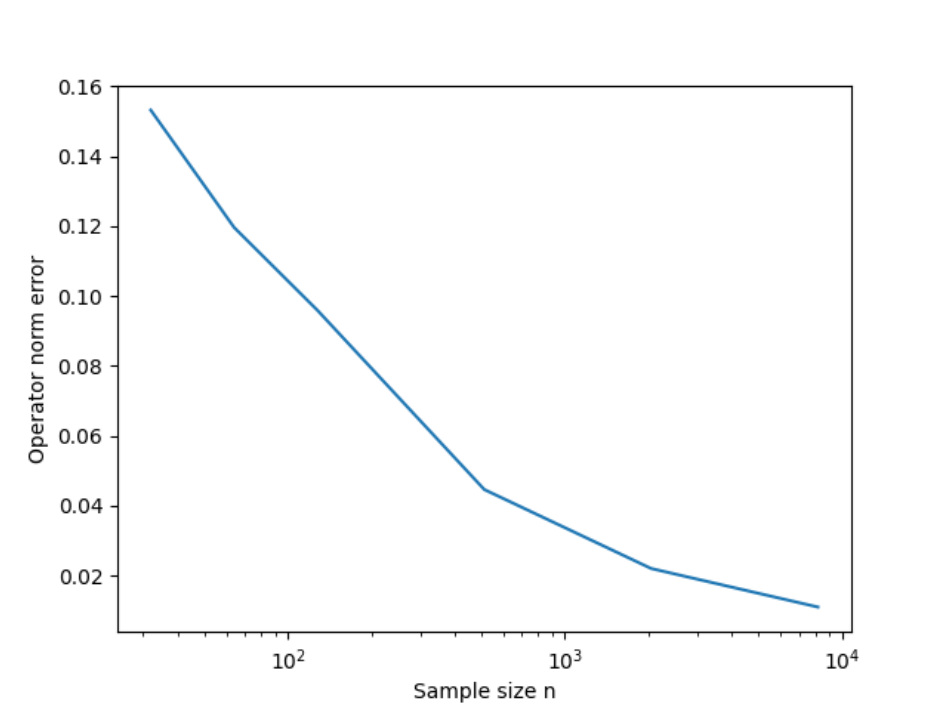


Figure 2: Behavior of the sample covariance matrix for estimating sequences of covariance matrices of increasing dimension, with fixed ratio $\alpha = 0.2$. Consistent with theoretical predictions, the operator norm error $\|\widehat{\mathbf{\Sigma}} - \mathbf{\Sigma}\|_2$ for this sequence decays at the rate $1/\sqrt{n}$

Covariance estimation - simulation

Solution: $\widetilde{\Sigma} := T_{\lambda_n}(\widehat{\Sigma})$ of the sample covariance, say with $\lambda_n = \sqrt{\frac{2 \log d}{n}}$

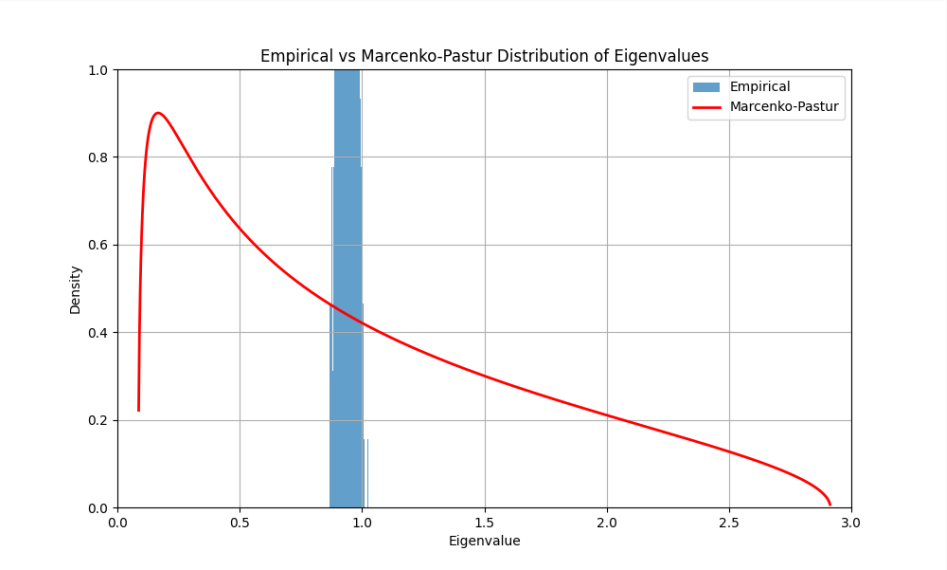


Figure 3: Behavior of the eigenspectrum $\gamma(\tilde{\Sigma})$ for a hard-thresholded version of the sample covariance matrix. Unlike the sample covariance matrix itself, it can be a consistent estimator of a sparse covariance matrix even for scalings such that $d/n = \alpha > 0$.

Summary

Covariance Estimation

- *Objective*: Estimate covariance matrix from i.i.d. samples.
- *Estimator*: Sample covariance matrix, evaluated with ℓ_2-operator norm.
- *Simulation*: Eigenvalue distribution compared to Marcenko-Pastur law.

Part 3
Nonparametric regression

Nonparametric regression

Objective:

- For an unknown distribution $\mathbb{P}_{X,Y}$ of (X, Y), we want to estimate $f(x) = \mathbb{E}[Y \mid x]$.

Nonparametric regression

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- Our goal is to find a function $\widehat{f}$ for which the mean-squared error (MSE) is as small as possible:

$$\widehat{f} = \operatorname*{argmin}_f \mathbb{E}_X \left[(f(X) - Y)^2 \right]$$

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Problem:

- This becomes extremely difficult in high dimensions! It is known as the *curse of dimensionality*.

Nonparametric regression

Suppose that the covariate vector X is uniformly distributed over $[0,1]^d$.

Sample size:

- To generate a good estimate of $\widehat{f}$, we need sufficient sample size.
- To ensure we have sufficient sample size, any vector $X' \in [0,1]^d$ should be relatively close to a least one of our samples. Consider:

$$\rho_\infty(n,d) := \mathbb{E}_{X',X}\left[\min_{i=1,\ldots,n} \|X' - X_i\|_\infty\right],$$

Problem:

- How many samples n do we need to collect as a function of the dimension d so as to ensure that $\rho_\infty(n,d)$ falls below some threshold δ?

Nonparametric regression

Theory: Actually, the lower bound

$$\log \rho_\infty(n, d) \geq \log \frac{d}{2(d+1)} - \frac{\log n}{d},$$

implies that a sample size $n > (1/\delta)^d$ is required to ensure that $\rho_\infty(n, d) \leq \delta$ holds, which implies exponentially large sample size.

Nonparametric regression

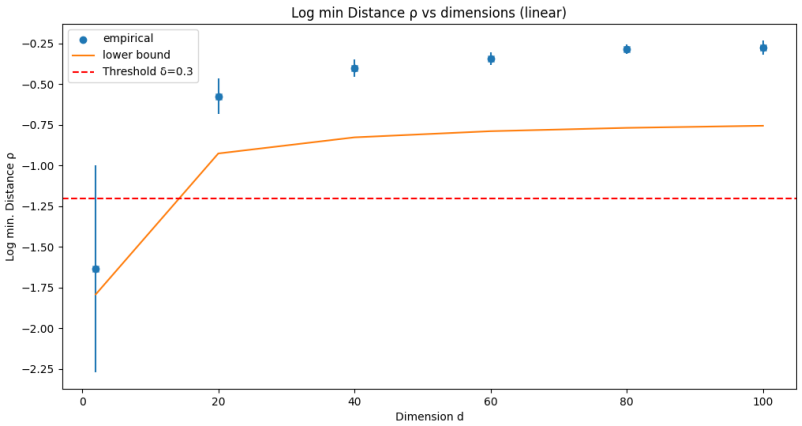
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Simulation We simulate $X \sim \mathcal{U}[0,1]^d, X' \sim \mathcal{U}[0,1]^d$ and we numerically estimate $\rho_\infty(n, d) = \mathbb{E}_{X',X}\left[\min_{i=1,\ldots,n} \|X' - X_i\|_\infty\right]$

Nonparametric regression - simulation



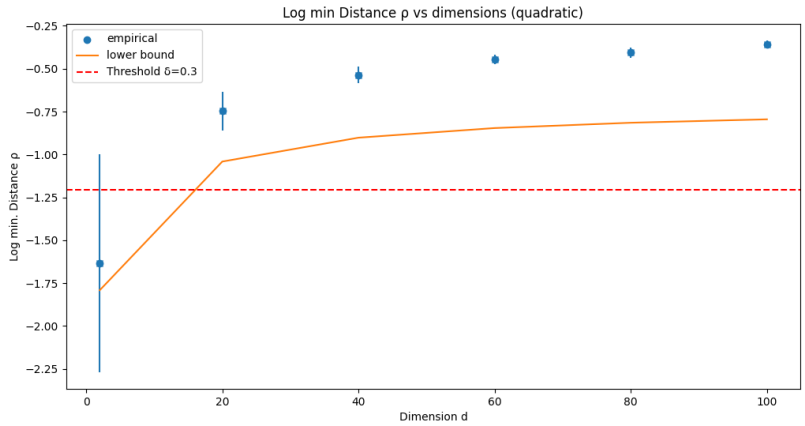


Figure 4: Behavior of the quantity $\rho_\infty(n, d)$ versus the dimension d, for different scalings of the pair (n, d). (a) Behavior of the variable $\rho_\infty(2d, d)$. (b) Behavior of the variable $\rho_\infty\left(d^2, d\right)$.

Nonparametric regression - solutions

Make additional assumptions such as:

- Additive decomposition

$$f(x_1, \ldots, x_d) = \sum_{j=1}^{d} g_j(x_j)$$

  where each univariate function $g_j(x_j) = \theta_j x_j$ for $\theta_j \in \mathbb{R}$, or g_j belongs to a RKHS.

- Sparse additive model (SPAM)

$$f(x_1, \ldots, x_d) = \sum_{j \in S} g_j(x_j).$$

- Multiple-index model

$$f(x_1, \ldots, x_d) = h(\mathbf{A}x),$$

  for some matrix $\mathbf{A} \in \mathbb{R}^{s \times d}$, and function $h : \mathbb{R}^s \to \mathbb{R}$.

Summary

Nonparametric Regression

- *Objective*: Estimate $f(x) = E[Y|x]$ to minimize mean-squared error.
- *Problem*: Curse of dimensionality makes high-dimensional estimation challenging.
- *Sample Size*: Exponentially large sample size needed to ensure closeness to any vector in high dimensions.

Thank you for your attention!